SimCert: A Classical-Simulability Audit of Trained Quantum-NLP Text Classifiers

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Abstract

Quantum natural language processing (QNLP) models, from compositional DisCoCat circuits to quantum self-attention networks and all-quantum token mixers, are promoted for a quantum benefit on text classification. That benefit is usually argued from accuracy, yet whether a *trained* QNLP circuit ever leaves the efficiently classically-simulable regime is rarely tested. We introduce SimCert, an open and model-agnostic protocol that audits this directly. It re-simulates each trained circuit as a bond-dimension-χ matrix product state, sweeps χ, and records the smallest value (written $\chi^\star$) at which a classical surrogate recovers the model's own predictions, together with an entanglement-removal ablation and two dequantization witnesses. We calibrate the criterion against a positive control whose required bond dimension is known in advance, and it recovers that number. Auditing five published QNLP architectures across four datasets and up to sixteen qubits, we find that a classical surrogate recovers the predictions at small and non-growing bond dimension. Attention-style models are recovered at χ=1, that is by a product state, and a compositional model at χ=2; $\chi^\star$ stays flat at 1 as the qubit count grows, for two architectures independently, while the exact bound grows to 256. The circuits do carry entanglement, but removing it barely moves the result. At the scales these models are tested, the quantum resource is therefore not the operative ingredient. This does not settle the asymptotic question, but it shifts the burden of proof: absent evidence that the required bond dimension grows with system size, the default reading of these models is classical. We release the audit as an open protocol and harness so that any QNLP model can be checked against this bar before a quantum benefit is claimed.

1 Introduction

Fault-tolerant quantum hardware does not yet exist, so most claims in quantum machine learning (QML) are established by classically simulating small circuits, and quantum NLP inherits the practice. A steady stream of QNLP models reports accuracy gains on text classification: compositional DisCoCat pipelines simulated for and run on real devices (Lorenz et al., 2021), quantum self-attention networks (Li et al., 2022; Chen et al., 2024), and, more recently, fully-quantum token mixers (Chen et al., 2025). The reported numbers are real. What is usually left implicit is the inference the reader is invited to draw, that the *quantum* structure is what does the work.

That inference is testable, and it is tested surprisingly rarely. Across the QNLP text-classification papers we examined (Appendix A), models are compared against classical *networks* of matched size but not against a classical *simulation of their own circuit*, which is the one baseline that would isolate whether the quantum resource, rather than the surrounding classical machinery or the trainable parameters, is responsible.[1] The same gap was recently closed for tabular QML (Bowles et al., 2024) and for quantum convolutional networks (Bermejo et al., 2024), resting on the theoretical footing that trainable variational models are constrained matrix product states (Shin et al., 2024) and that the absence of barren plateaus tends to imply classical

[1] We count a paper as testing simulability only if it reports a classical surrogate of the trained circuit, meaning a tensor-network contraction, a Pauli-propagation estimate, or an explicit entanglement-removal ablation, rather than merely a separately-trained classical baseline. This is a focused review, not a saturated systematic survey; scope and counts are in Appendix A.

simulability (Cerezo et al., 2025). So which QNLP results survive a classical surrogate of the trained circuit? That is the question we take up.

This paper makes three contributions, each an audit rather than a new model. First, we define a scoped simulability criterion for trained QNLP circuits, namely the smallest MPS bond dimension $\chi^\star$ at which a classical surrogate recovers the model's predictions, and we package it with an entanglement-removal ablation, the entanglement entropy, a geometric-difference kernel test, and a data-reuploading spectral test into a per-model simulability certificate. Second, we audit five published QNLP architectures, namely DisCoCat as realised in `lambeq`, the quantum self-attention models `QSANN` and `QMSAN`, the all-quantum token mixer `CLAQS`, and a variational-classifier reference, across four datasets, and we report how $\chi^\star$ scales with system size. Third, we release an open, model-agnostic harness that consumes any trained circuit through a common intermediate representation, so the certificate can be recomputed and extended by others.[2]

The paper front-loads its epistemics. Section 2 places the work among the audits it extends. Section 3 sets out what a simulability audit can and cannot establish. Section 4 fixes definitions and the certificate. Sections 5 and 6 describe the audited models, the data, and the surrogate. Section 7 reports the findings, and Section 8 states the caveats and the open questions we think follow.

2 Related work

Our closest relatives are recent audits of quantum machine learning in other domains. Bowles et al. (2024) benchmark twelve QML models across many tabular datasets, match them against out-of-the-box classical baselines, and find that removing entanglement rarely hurts; we adopt their entanglement-removal ablation and their discipline of scoping every claim to the tested regime. Bermejo et al. (2024) show that trained quantum convolutional networks are effectively classically simulable by propagating low-weight observables, the vision-domain analogue of the question we ask for text. The theoretical footing is that a trained variational model is a constrained matrix product state (Shin et al., 2024), and that the very structure making such models trainable, the absence of barren plateaus, tends to make them classically simulable (Cerezo et al., 2025); our bond-dimension criterion is the operational instrument these results invite. Our two auxiliary witnesses come from the same program: the geometric difference between quantum and classical kernels (Huang et al., 2021) and the Fourier spectrum accessible to a data-reuploading encoding (Schuld et al., 2021).

This program has a longer arc. Tang (2019) showed that a claimed exponential quantum speedup for recommendation systems dissolved once the classical algorithm was given comparable sampling access, establishing dequantization as a routine obligation rather than an exotic outcome. Schreiber et al. (2023) make that obligation concrete for learning models by constructing classical surrogates that reproduce a trained quantum model's input-output behaviour, and Huang et al. (2020) show that few measurements suffice to build a classical description adequate for predicting many properties of a state. Our surrogate is of the same species, specialised to a bond-dimension-limited MPS so that the knob controlling classical power is explicit and sweepable.

The trainability literature supplies the other half of the motivation. Barren plateaus make deep random circuits untrainable (McClean et al., 2018), which is why practical variational models (Peruzzo et al., 2014; Cerezo et al., 2021) adopt shallow, structured ansatze. But the structures that avoid plateaus, such as quantum convolutional networks (Cong et al., 2019; Pesah et al., 2021), are precisely those Cerezo et al. (2025) identify as classically simulable, and even plateau-free shallow models are beset by poor local minima (Anschuetz and Kiani, 2022). Larocca et al. (2025) survey where this leaves the field. The tension is structural rather than incidental, and our audit measures which side of it a trained QNLP circuit falls on.

The QNLP models we audit inherit a specific lineage. Compositional distributional semantics assigns a sentence a meaning built from its pregroup parse (Coecke et al., 2010); Zeng and Coecke (2016) observed that this structure maps onto quantum circuits, Meichanetzidis et al. (2021) built the near-term pipeline, and Kartsaklis et al. (2021) packaged it as `lambeq`, which our compositional producer uses. The line continues onto hardware (Duneau et al., 2024), and Nausheen et al. (2025) survey the wider field. The attention-

[2] Anonymized for review: `https://anonymous.4open.science/r/simcert`.

based models instead import the classical transformer's mechanism (Vaswani et al., 2017), whose pretrained descendants (Devlin et al., 2019) are the baselines QNLP accuracy claims are implicitly measured against.

On the QNLP side, the models we audit are the compositional DisCoCat framework (Lorenz et al., 2021), the Gaussian-projected and mixed-state quantum self-attention networks (Li et al., 2022; Chen et al., 2024), and the all-quantum token mixer (Chen et al., 2025). Each reports an accuracy comparison against a classical network, and none reports a classical surrogate of its own trained circuit, which is the gap we fill. What is new here is not the idea of a simulability audit but its first application to language, where the circuits are structurally unlike the tabular and vision cases (compositional cups, attention, and mixed states), and its packaging as a reusable protocol and harness rather than a one-off study. We read our results as extending the tabular and vision audits to language, and as agreeing with them, rather than as a challenge to any individual QNLP paper.

3 What a simulability audit can and cannot tell us

A skeptical result earns trust only if it anticipates how it might be wrong, so we state the limits before the evidence. An audit of this kind speaks to *trained* models at the scales they are actually run. It does not, and cannot, prove that no QNLP model at any size resists classical simulation, because the interesting asymptotic question is exactly the one simulation cannot reach. What it can do is check a concrete and load-bearing assumption, that the accuracy a model reports depends on quantum structure a cheap classical surrogate cannot reproduce.

Three failure modes shape the design. A surrogate that is too weak will fail to recover predictions for uninteresting numerical reasons, so we sweep the surrogate's power, the bond dimension, rather than fixing it, and report the whole curve. Recovering accuracy and recovering the model's function are different tests, since a surrogate could match the label while computing a different function, so we report agreement with the trained model's own predictions and not only accuracy against ground truth. Finally, the absence of a quantum effect on a saturated benchmark says little, so we include a genuinely hard task and are explicit about where a benchmark is degenerate. None of this settles the asymptotic question. It settles the empirical one that current QNLP papers implicitly answer in the affirmative.

4 Framework and definitions

Setup. A QNLP text classifier maps an input x, a sentence or a sequence of token embeddings, to a trained circuit that prepares a state $|\psi_\theta(x)\rangle$ on n qubits, from which a readout, a set of Pauli expectation values possibly followed by a small classical head, produces a label $\hat{y}_\theta(x)$. We treat the trained parameters θ as fixed and ask about the function $\hat{y}_\theta$ (Figure 1).

Which notion of simulability. We use the standard hierarchy. A model is classically simulable (CSIM) if its outputs can be computed classically in polynomial time, and classically simulable up to a data-loading step, a weaker and common-in-practice notion, if the only obstruction is preparing the input state (Cerezo et al., 2025; Shin et al., 2024). Our surrogate is an MPS, which realises CSIM whenever the required bond dimension is small, and we make no claim about regimes where it is not. Every claim below has the form "simulable at bond dimension up to χ, at these model sizes, on these datasets" (Figure 2).

The bond-dimension criterion. For a trained state we form its optimal bond-dimension-χ MPS approximation $|\psi^{(\chi)}\rangle$ by sequential singular-value truncation, recompute the readout, and obtain a truncated prediction $\hat{y}^{(\chi)}(x)$ (Figure 4). Over the test set we report the truncated accuracy $A(\chi)$, the agreement $\pi(\chi) = \Pr_x[\hat{y}^{(\chi)}(x) = \hat{y}(x)]$ with the untruncated model, and the mean state fidelity $F(\chi)$. The certificate's headline scalar is

$$\chi^\star \;=\; \min\{\chi : A_{\text{full}} - A(\chi) \leq \tau\}, \tag{1}$$

the smallest bond dimension whose accuracy is within a tolerance τ of the untruncated model. The choice of bond dimension as the knob is not arbitrary. Matrix product states arose as the variational class behind the density-matrix renormalization group (White, 1992; Schollwöck, 2011; Orús, 2014) and provably represent

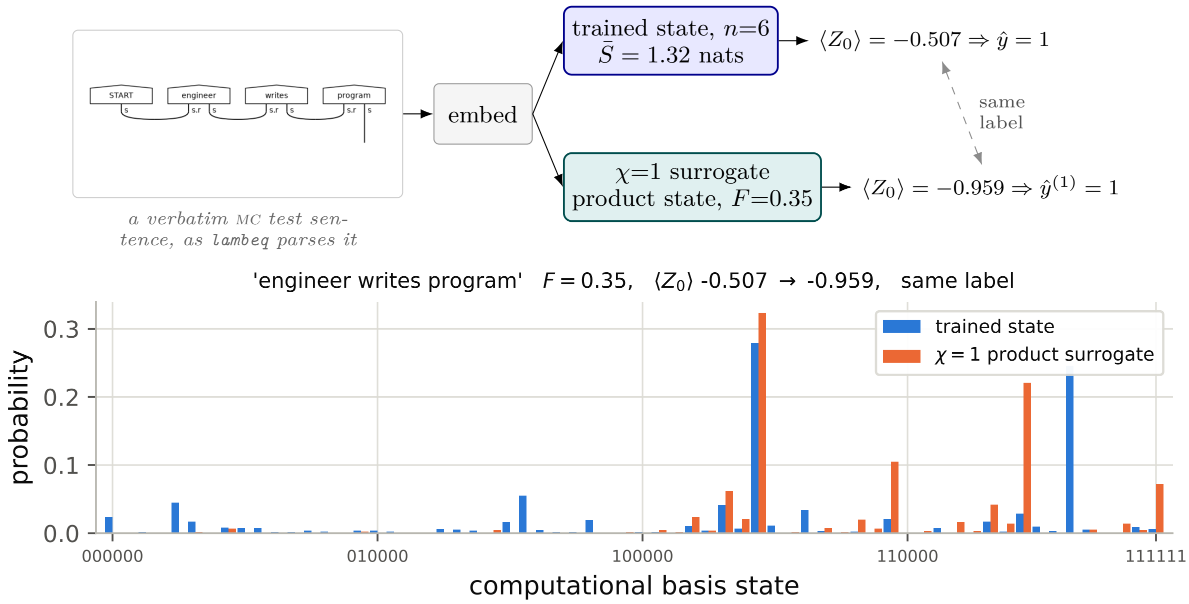


Figure 1: The audit on one real example, showing the object it operates on. Top, the sentence `engineer writes program`, a verbatim MC test item, drawn as the compositional parse `lambeq` builds for it; the curved wires are the grammar cups that make the sentence's meaning a contraction rather than a product. The trained reference classifier prepares a six-qubit state from it, and SimCert prepares the best bond-dimension-one approximation of that state. Bottom, the two states themselves. They are plainly different, overlapping by only $F = 0.35$, and the surrogate is a product state carrying no entanglement where the trained state carries 1.32 nats. They nonetheless return the same label, because the readout $\langle Z_0 \rangle$ stays on the same side of the decision boundary. That gap, between a state that is visibly not reproduced and a decision that is, is what this paper measures. All values are measured on the audited configuration at seed 1, and the trace reproduces that run's stored certificate exactly.

gapped one-dimensional ground states at a bond dimension that does not grow with system size (Verstraete and Cirac, 2006). Vidal (2003) turned this into the statement we rely on: a quantum computation whose entanglement stays bounded can be simulated classically at a cost polynomial in the system size and exponential only in the entanglement, so bond dimension is the natural currency in which to price a circuit's classical difficulty. The same class is a capable learner in its own right (Stoudenmire and Schwab, 2016), which is why a low-χ surrogate that matches a quantum model is evidence about the model rather than about the surrogate's weakness. We set τ to the model's own generalization gap, since differences smaller than the gap the model already exhibits are not meaningfully quantum resource, and we report robustness to two other tolerances in Appendix D. A model whose $\chi^\star$ is small and does not grow with n is, in this operational sense, recovered by a cheap classical surrogate; we call such a model "bond-cheap" and use the term plainly thereafter.

Result (informal). *For every audited model at the tested scales, $\chi^\star$ is a small constant, one or two, and does not grow when the qubit count is increased on a fixed task, while the bond dimension required for an exact MPS grows as $2^{\lfloor n/2 \rfloor}$.*

Corroborating witnesses. Bond dimension is one lens. The certificate adds three inexpensive ones. The first is an entanglement-removal ablation that retrains a product-state twin of the model and reports $\Delta A_{\text{ent}} = A_{\text{full}} - A_{\text{sep}}$, following Bowles et al. (2024). The second is the mean bipartite entanglement entropy $\bar{S}$ of the trained states, which lower-bounds the χ an exact representation needs. The third pairs two dequantization probes where they apply, the geometric difference g_{CQ} between the trained quantum kernel and a classical kernel (Huang et al., 2021) and the effective data-reuploading Fourier degree (Schuld et al., 2021). Both probes come from the kernel reading of quantum models, under which a data encoding is a

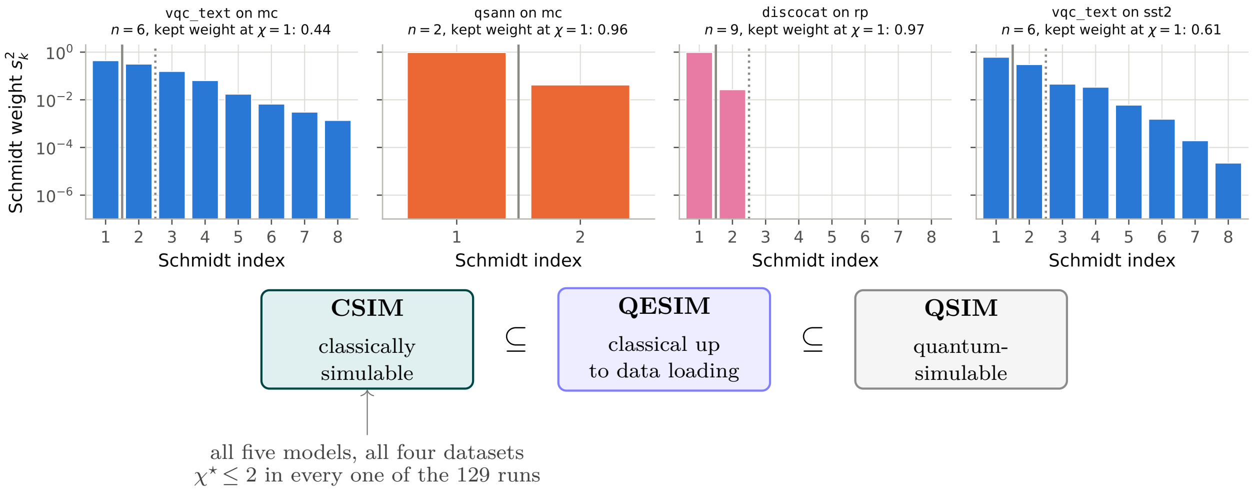


Figure 2: The states that decide the verdict, and the class the verdict names. Top, the Schmidt spectrum at the central cut of one trained state per audited case, measured on the audited configurations. The solid rule marks where a χ=1 surrogate truncates and the dotted rule where χ=2 does, so each panel shows directly how much of the state is being discarded. The compositional `discocat` is the informative one: its spectrum carries two weights and then falls off a cliff, so it is almost exactly rank two, which is why it is the one model in the paper that needs χ=2 and cannot be recovered by a product state. The attention model is near rank one already. The reference classifier keeps only 0.44 of its weight in the leading Schmidt value on MC and 0.61 on SST-2, so its state is genuinely spread, and yet its decisions survive truncation to a product state, which is the dissociation this paper reports. Weights below 10^{-7} are clipped so the `discocat` cliff does not flatten the other panels. Bottom, the nested hierarchy: a classically simulable model (CSIM) is a special case of one simulable up to a data-loading step (QESIM), itself a special case of quantum-simulable (QSIM) (Cerezo et al., 2025; Shin et al., 2024). SimCert certifies membership in CSIM whenever the trained state truncates to a small bond dimension, and we make no claim outside that innermost class.

feature map (Havlíček et al., 2019; Schuld and Killoran, 2019) and a trained variational model is equivalent to a kernel machine (Schuld, 2021); on that reading the encoding, not the ansatz, is where a separation would have to live, and provable separations are known only for carefully constructed problems (Liu et al., 2021). We report the full vector rather than collapsing to a single verdict, because, as Section 7 shows, the witnesses can legitimately disagree.

5 Models and datasets audited

Selection. We audit published QNLP text classifiers that target sentence or text classification, fit within a simulator's reach at their published qubit counts, and can be reimplemented from their descriptions. Applying these criteria to the QNLP literature leaves a compact set spanning the two dominant paradigms, compositional and attention-based, plus a recent all-quantum mixer, and we add one plain variational classifier as a reference point. Table 1 compares them at a glance and stands in for a wall of per-model circuit diagrams; representative ansätze appear in Figure 3 and Appendix B. All four use parameterised circuits of the kind now standard in quantum machine learning (Biamonte et al., 2017; Mitarai et al., 2018; Benedetti et al., 2019). Such circuits are usually characterised by ansatz-level descriptors, expressibility and entangling capability (Sim et al., 2019), effective dimension (Abbas et al., 2021), or generalization bounds in the number of trainable gates (Caro et al., 2022). Those quantities describe what an ansatz could represent. We deliberately measure something else, what a trained instance turns out to need, because the two can differ and it is the second that decides whether a reported accuracy rests on quantum structure. None of the attention or mixer models ship official code, so we reimplement from the equations and report the reproduction gap openly in Section 7, while the compositional model uses the official `lambeq` pipeline.

Model	Paradigm	Cross-token mixing	Readout	State
`discocat`	compositional	grammar cups (post-selection)	open-wire Born rule	pure
`qsann`	attention	Gaussian of measured Q/K (classical)	Pauli vector to head	pure
`qmsan`	attention	$\mathrm{tr}(\rho_q \sigma_k)$ (mixed-state)	Pauli vector to head	mixed
`claqs`	token mixer	complex LCU and QSVT polynomial	XYZ Paulis to MLP	pure
`vqc_text`	reference	entangling ring	$\langle Z_0 \rangle$	pure

Table 1: The five audited models, one row per architecture, grouped by paradigm. `qmsan` is the only mixed-state model and is audited through the purification of its reduced density matrices. Full circuits, gate sets, and parameter counts are in Appendix B.

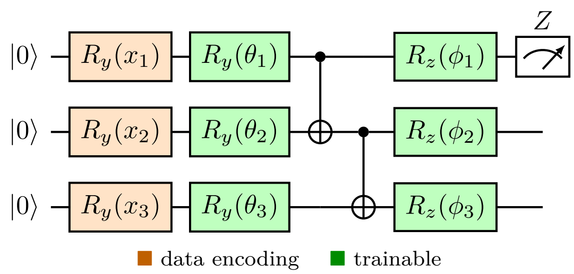


Figure 3: A representative variational ansatz (the reference classifier; the attention and mixer models use the same primitives). Each input feature is angle-encoded ($R_y(x_i)$, orange), followed by a trainable layer ($R_y(\theta_i)$, green), a CNOT entangling chain, and a trainable $R_z(\phi_i)$. The block repeats L times with the data re-uploaded each layer, and the class is read from $\langle Z_0 \rangle$. The audit forms the bond-dimension-χ MPS approximation of the state prepared *just before* measurement. Per-model circuits are in Appendix B.

Datasets. We use the tasks these models were built on. MC, meaning classification of cooking versus computing, and RP, a relative-pronoun task, are the canonical DisCoCat benchmarks of Lorenz et al. (2021). MC is small and near-separable, whereas RP is the harder, low-data task where published accuracies actually differ across models. We add a balanced subset of SST-2, the sentence-level binary split of the Stanford Sentiment Treebank (Socher et al., 2013), as a genuinely non-separable sentiment task, and a synthetic MC-style set for controlled scaling. Sizes and preprocessing are in Appendix C. Two limitations are worth stating here rather than burying. Our RP reimplementations use our own token embeddings, and our DisCoCat pipeline uses an offline reader in place of the trained grammar parser whose model host is no longer reachable. Both cost accuracy on RP, as we report.

6 The classical surrogate and simulation protocol

We do not conjecture simulability, we realise it. Each trained model exports its per-example circuit to a common representation, OpenQASM for gate circuits, or, for the token mixer whose state is a matrix polynomial rather than a gate sequence, the trained statevector directly. Statevector simulation and gradients go through PennyLane (Bergholm et al., 2018), and every model is small enough that the whole audit runs on a laptop CPU, which is the regime these architectures are published in (Preskill, 2018). A single audit consumer then re-simulates every model identically (Algorithm 1), so the audit never depends on which framework produced a circuit. The MPS truncation itself is an exact sequential-SVD computation, cross-checked against an independent tensor-network simulator. On circuits with known answers, a GHZ state and a product state, it reproduces the analytic bond structure, which is our correctness check for the whole pipeline.

The surrogate's cost is explicit. An MPS sweep on an n-qubit, depth-d circuit scales as $\mathcal{O}(n\, d\, \chi^2)$, and the exact-representation bound is $\chi = 2^{\lfloor n/2 \rfloor}$, so a small measured $\chi^\star$ against a rapidly growing exact bound is precisely the signal we are after. All experiments run on a single CPU with no GPU. Where a

Algorithm 1 SimCert bond-dimension audit for one trained model

input: trained model M, test set D, tolerance τ, bond dimensions X
for each example $x \in D$ **do**
  export the trained units of x (one whole-sentence circuit, or per-token circuits)
  for each $\chi \in X$ **do**
    truncate each unit's state to bond dimension χ; recompute its Pauli readout
    $\hat{y}^{(\chi)}(x) \leftarrow$ model's own decision rule on the truncated readout
  end for
end for
$A(\chi), \pi(\chi), F(\chi) \leftarrow$ accuracy, agreement, fidelity aggregated over D
$\chi^\star \leftarrow \min\{\chi : A_{\text{full}} - A(\chi) \leq \tau\}$ ▷ Eq. 1
return certificate ($\chi^\star$, ΔA_{ent}, $\bar{S}$, g_{CQ}, Fourier degree)

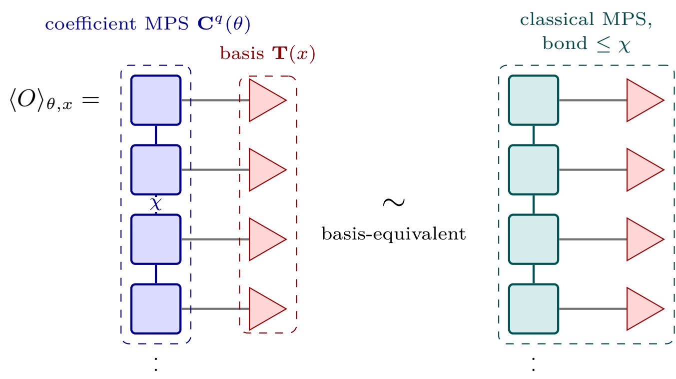


Figure 4: Following the dequantization picture of Shin et al. (2024), the trained model's expectation value is a *coefficient* matrix product state $\mathbf{C}^q(\theta)$ (blue) contracted with a fixed set of basis functions $\mathbf{T}(x)$ (red), and its bond dimension caps the entanglement it can carry. If that bond dimension is small, a classical MPS of the same size (teal) contracts with the same basis functions to reproduce the output, so the model is basis-equivalent to a cheap classical one. SimCert measures this bond dimension directly: it truncates the trained state to bond dimension χ and finds the smallest χ that still recovers the predictions, which for every audited model is 1 or 2, against an exact-representation bound of $2^{\lfloor n/2 \rfloor}$.

reimplementation is heavier than the hardware comfortably allows we say so and reduce scale rather than quietly dropping runs, and full configurations are in Appendix D.

7 Results

We report three findings, then a subsection of effects we looked for and did not see. Unless noted, numbers are the mean over seeds, with the per-row seed count in Table 2; the aggregate certificate is Table 2 and the reproduction gap is Table 3. We quantify uncertainty in two ways that do not rely on the seed count alone. For every truncated surrogate we compute a 95% example-level bootstrap confidence interval on its retained accuracy, and we compare it against the full model with an exact paired McNemar test (McNemar, 1947) on the test predictions, the paired form being the appropriate one when two models are scored on the same examples (Dietterich, 1998); the intervals are percentile bootstrap intervals (Efron, 1979). A large McNemar p-value means the surrogate is not significantly different from the full model, which is the simulability signal; a small p-value flags a surrogate that is significantly worse. Wherever the attention and mixer models saturate, the product-state surrogate does not merely fail to differ significantly from the full model, it reproduces every test prediction, so the contingency table is empty and the comparison is decided by $d^{\chi=1}$=0 rather than by a p-value. On the harder real-data tasks the surrogate does flip a few decisions, on average between 0.5 and 5 test examples, but never significantly, with p-values from 0.375 to 0.981.

The reference model `vqc_text` reaches $d^{\chi=1}$=0 on all three of its datasets. The one model whose decisions truncation to χ=1 does change is `discocat`, where the surrogate is significantly worse on MC (p=0.014) and sits at the conventional boundary on MC-REAL (p=0.050), which is exactly the pair of tasks whose certificate reports $\chi^\star$=2. On RP that same comparison is not significant (p=0.318), which is a statement about the task rather than about the model: `discocat` sits close to chance on RP (Table 3), so there is little decision content left for truncation to destroy (Table 2).

7.1 A classical surrogate recovers the predictions at small bond dimension

Across models and datasets, a bond-dimension-one or bond-dimension-two surrogate recovers the trained model's predictions (Figure 6). On the canonical MC task the attention and mixer models `qsann`, `qmsan`, and `claqs`, together with the reference classifier, all reach full accuracy at χ=1, a product state, with no accuracy lost; their curves in Figure 6 are flat down to the smallest bond dimension, a product state. The compositional `discocat` is the exception and needs χ=2, but no more. The pattern holds on real data. Our DisCoCat run reproduces the published MC accuracy to within 0.02, at 0.778 against 0.798 (Table 3), and is recovered at χ=2. The scales are already low, so in the operational sense of Section 4 these trained models are bond-cheap.

7.2 $\chi^\star$ does not grow with system size

The obvious objection is that small circuits are trivially simulable, so the audit only bites at scale. Figure 5 tests this directly. On a fixed task we increase the qubit count and track $\chi^\star$, for the reference model from four to sixteen qubits and independently for `qsann`, whose per-token attention circuits are structurally unlike the reference ansatz, from four to ten. For both, $\chi^\star$ stays flat at 1 while the exact bond bound grows from 4 to 256. What is exactly true at every n, and what the $\chi^\star$=1 verdict rests on, is that the product-state surrogate reproduces the untruncated accuracy; the retained-accuracy curves are not monotone in χ, and dip by one or two test examples at intermediate bond dimension before returning. Truncation error and decision error are different quantities, and only the second defines $\chi^\star$. Adding qubits does not raise the bond dimension a classical surrogate needs, and at sixteen qubits a product state still reproduces the predictions that an exact matrix product state would need bond dimension 256 to represent. A flat $\chi^\star$ would be uninteresting if it simply tracked a classifier decaying toward chance, but neither model decays: the reference model fluctuates between 0.52 and 0.66 with no trend in n, reaching its highest accuracy at sixteen qubits, and `qsann` holds between 0.62 and 0.66. *The gap between what the model uses and what it could use widens with every qubit.*

7.3 Removing the quantum resource barely moves the result

Two further witnesses agree (Figure 7). The trained circuits are not trivial product states, since their mean bipartite entanglement entropy runs from about 0.2 to 1.0 nats on MC (Figure 7a) and from about 0.1 to 1.3 nats across all datasets (Table 2), yet truncating to χ=1 loses no predictions for the attention models (Figure 7b), so the entanglement they build is not load-bearing for their decisions. The entanglement-removal ablation tells the same story on the easy tasks, where a retrained product-state twin matches the full model and ΔA_{ent} is within noise of zero for `qsann` and `qmsan` on MC. The one place the state fidelity climbs gradually with χ is the eight-qubit mixer `claqs`, where fidelity rises from 0.72 at χ=1 to 0.99 at χ=4; this is also the one place the bond-dimension axis has room to resolve structure, and even there the predictions are recovered at χ=1.

7.4 A positive control: $\chi^\star$ tracks a dialled entanglement resource

A reader is entitled to ask whether the audit is capable of reporting anything other than $\chi^\star$=1. A uniform verdict is only informative if the instrument would have said otherwise had the circuits been different. We therefore calibrate it against a family of states whose required bond dimension is known in advance, and audit them with the same code path, tolerance rule and grid used for every model above.

The family has one integer knob. On n=$2m$ qubits, pair j joins wire j and wire j+m; the first k pairs are entangled and the remaining $m-k$ are product states. The half-way cut is straddled by exactly the k

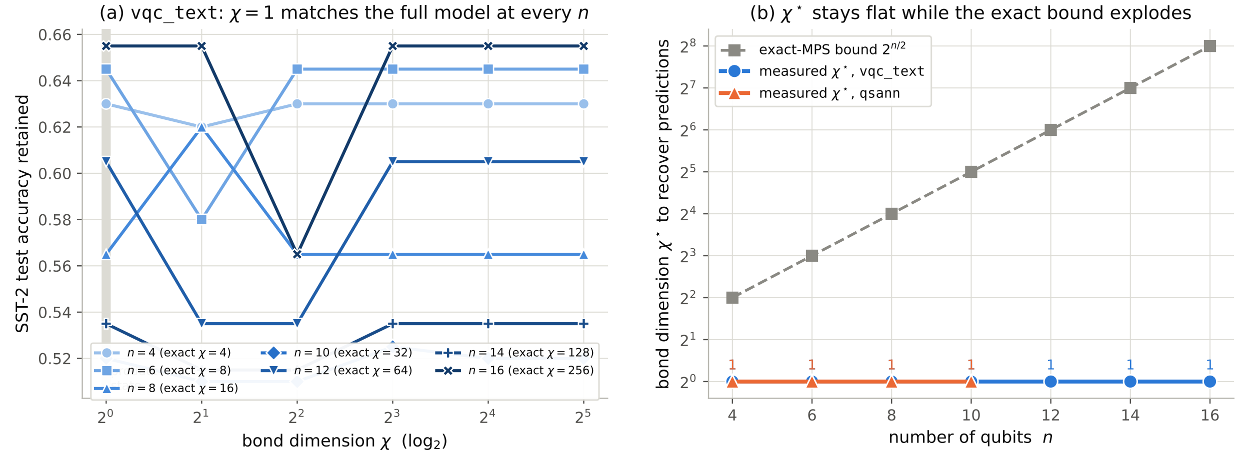


Figure 5: The required bond dimension $\chi^\star$ does not grow with qubit count. Left, accuracy retained versus bond dimension for the reference model at $n \in \{4, \ldots, 16\}$ on SST-2; at every n the χ=1 point equals the untruncated accuracy, with small non-monotone excursions in between. Colour runs light to dark with n, since n is an ordered quantity. Right, the measured $\chi^\star$ for the reference model and for `qsann` both stay at 1 while the exact-MPS bound $2^{n/2}$ (grey, dashed) grows exponentially on the log axis, reaching 256 at sixteen qubits. A product-state surrogate recovers the predictions at every size tested, for two architectures that share no circuit structure. Accuracy is modest throughout but shows no downward trend in n, so the flat $\chi^\star$ is not an artifact of a classifier decaying to chance.

entangled pairs, so the Schmidt rank there is 2^k and no matrix product state of bond dimension below 2^k can represent the state. Turning k moves physical resource and nothing else: the qubit count, the wire order, the readout, its weight, and the distribution of the full model's own readout values are all held fixed. The decision reads the m straddling-pair correlators $\langle X_j X_{j+m} \rangle$ and returns their parity. Parity makes every pair load-bearing, because a single destroyed correlator collapses the product; and an X-correlator is the right probe, since a diagonal $\langle Z_j Z_{j+m} \rangle$ is reproduced exactly by a single retained branch and is therefore blind to entanglement. Both branches of the construction give the same correlator value, so at every k the full model computes the same function of the same readout, and only the entanglement behind it changes.

Figure 8 reports the result. $\chi^\star$ rises from 1 at k=0 to 16 at k=4, spanning the whole grid, and the measured value never exceeds the exact bound 2^k, meeting it at four of the five points. Accuracy sits exactly at chance below threshold and exactly at 1.0 at and above it, so each point is a sharp step rather than a statistical estimate. Two checks fix the endpoints: a separable member (k=0, entropy 0) is certified at $\chi^\star$=1, and a maximally entangled member (k=m, entropy $m \ln 2$, product-state fidelity 2^{-m}) is not. The one point below the bound is k=3, where $\chi^\star$=4 against a bound of 8: there the truncated correlators are attenuated but still sign-correct, so the parity survives at half fidelity. That is the same decision-versus-state distinction that runs through this paper, visible here against a known ground truth.

The control therefore does two things. It shows the criterion is sensitive, and it places the audited models on the curve: they sit at the k=0 end, where a product state suffices, and not merely at a point where the instrument is blind.

7.5 Effects we did *not* observe

We looked for, and did not find, three things. We did not see $\chi^\star$ grow with qubit count on a fixed task (Figure 5). We did not see the entanglement-removal ablation cost accuracy on the tasks where the models saturate. And we did not see the compositional model behave like the attention models. `discocat` alone needs χ=2, and truncating it to χ=1 collapses its accuracy toward chance, so its grammar cups create entanglement that is, minimally, load-bearing. This lone exception matters more than the uniform cases, because it shows the audit is not simply reporting χ=1 by construction. The two verdict classes also separate

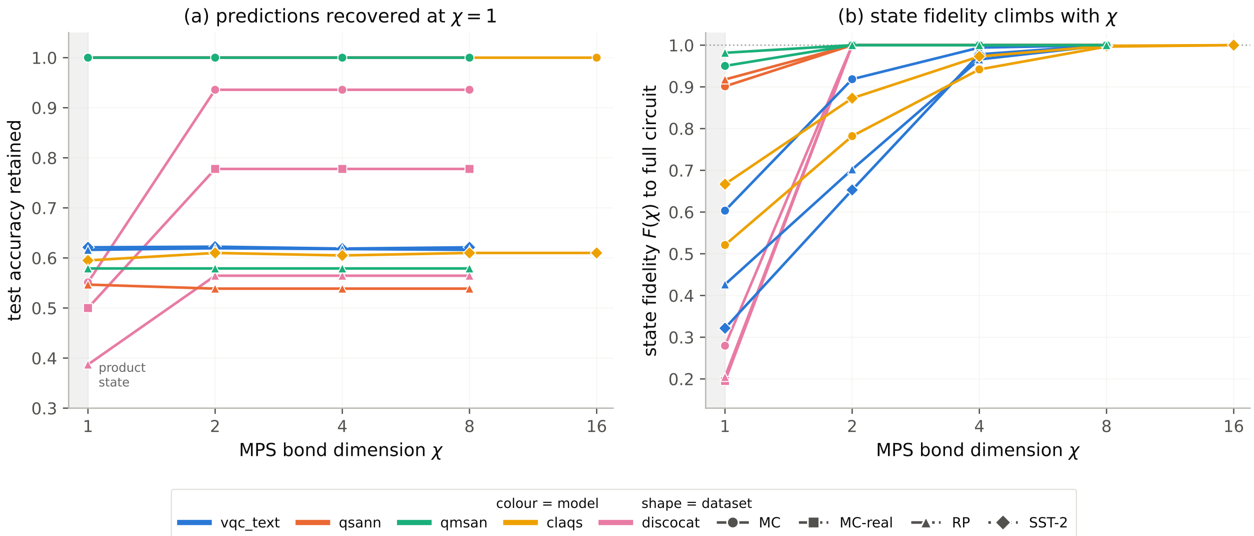


Figure 6: Two views of the same bond-dimension sweep, one curve per model and dataset, averaged over seeds. Color encodes the model and marker shape encodes the dataset, so each series stays distinguishable in grayscale. Panel (a) plots test accuracy retained against χ. A curve that is flat down to χ=1 is the signature of a bond-cheap decision, since the classical surrogate loses no predictions as it is weakened to a product state. The only curves that climb are the three compositional `discocat` runs, which recover their accuracy at χ=2; every other series is flat. Panel (b) plots the state fidelity $F(\chi)$ between the truncated and the full circuit for the same runs. This climbs from as low as 0.2 at χ=1 toward one, so the trained states do carry entanglement. Read together the panels reconcile the finding: the wavefunctions are entangled, yet the decisions built on them do not need that entanglement at the sizes tested.

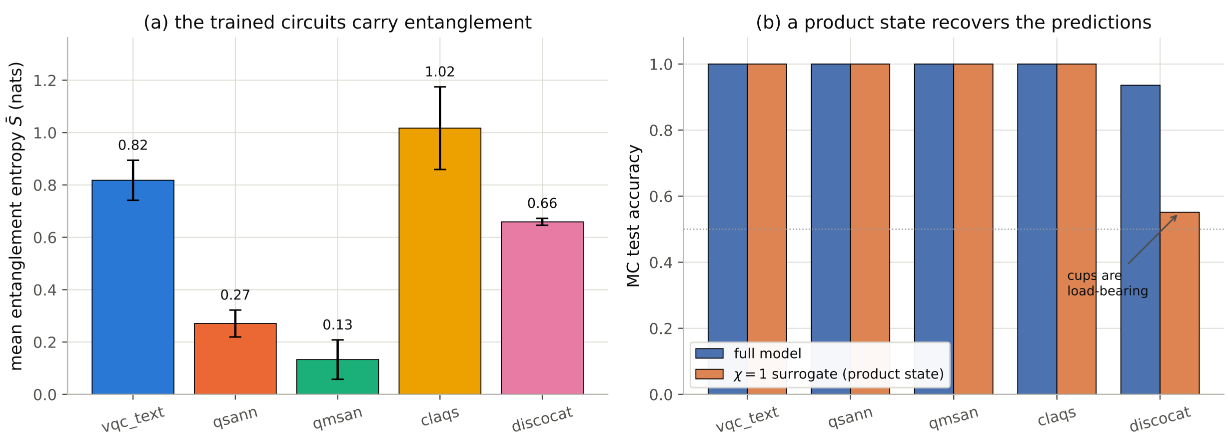


Figure 7: Entanglement is present but, with one exception, not load-bearing, on MC for all five models). Left, the mean bipartite entanglement entropy $\bar{S}$ is non-zero for every model, so the trained states are genuinely entangled. Right, the full model and its χ=1 product-state surrogate reach the same accuracy for the attention and mixer models, whereas for `discocat` the χ=1 surrogate drops toward chance, marking the one place where the entanglement built by its grammar cups is load-bearing.

quantitatively, and in the direction they should: over every audited run, the product surrogate's mean state fidelity is 0.66 where the certificate reports $\chi^{\star}$=1 and 0.22 where it reports $\chi^{\star}$=2. The criterion fires where a product state is genuinely far from the trained one. The positive control of Section 7.4 makes the same point against a known ground truth, by dialling the entangled resource and recovering the bond dimension it implies.

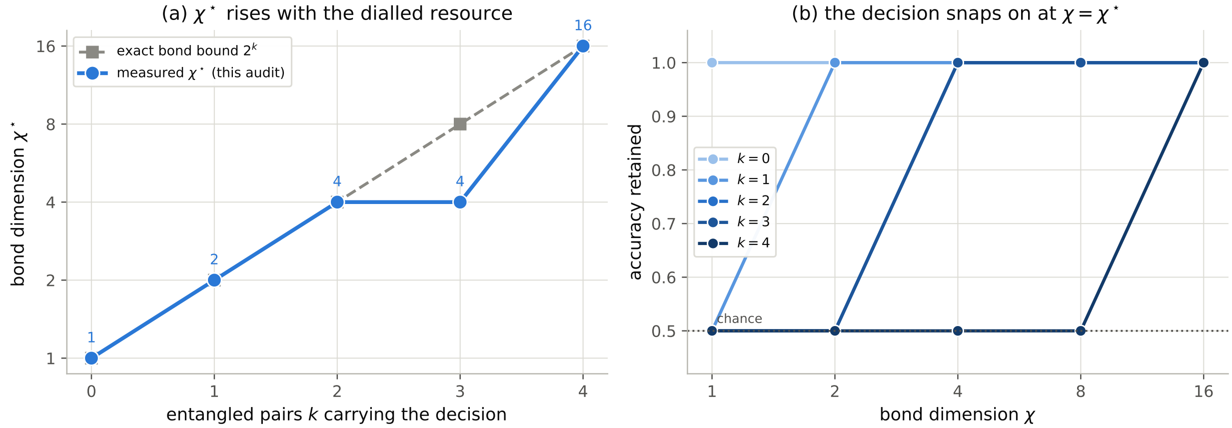


Figure 8: The audit discriminates. Left, measured $\chi^\star$ against the number of entangled pairs k carrying the decision, with the exact bond bound 2^k dashed; the two coincide except at k=3. Right, retained accuracy against χ for each k: the decision sits at chance until χ reaches $\chi^\star$, then snaps to 1.0. Qubit count, wire order, readout and readout weight are identical at every k, so the only thing varying along these curves is the entanglement the decision depends on.

Model	Data	seeds	Full acc	Acc@χ=1	$p^{\chi=1}_{\mathrm{McN}}$	$d^{\chi=1}$	$\chi^\star$	$\bar{S}$ (nats)	ΔA_{ent}
`claqs`	mc	3	1.000±0.000	1.000±0.000	1.000	0	1	1.017±0.158	n/a
`claqs`	sst2	1	0.610	0.595	0.375	5.0	1	0.723	n/a
`discocat`	mc	3	0.936±0.065	0.551±0.079	0.014	11.3	2	0.659±0.013	n/a
`discocat`	mc_real	3	0.778±0.087	0.500±0.000	0.050	11.0	2	0.667±0.019	n/a
`discocat`	rp	20	0.565±0.098	0.387±0.000	0.318	13.0	1/2	0.672±0.018	n/a
`qmsan`	mc	8	1.000±0.000	1.000±0.000	1.000	0	1	0.133±0.075	0.000±0.000
`qmsan`	rp	20	0.579±0.063	0.579±0.059	0.938	0.5	1	0.051±0.017	0.011±0.026
`qsann`	mc	8	1.000±0.000	1.000±0.000	1.000	0	1	0.271±0.051	0.000±0.000
`qsann`	rp	20	0.539±0.085	0.547±0.094	0.981	0.7	1	0.232±0.045	-0.024±0.043
`vqc_text`	mc	8	1.000±0.000	1.000±0.000	1.000	0	1	0.818±0.076	0.168±0.186
`vqc_text`	rp	20	0.616±0.071	0.616±0.071	1.000	0	1	1.232±0.140	0.089±0.118
`vqc_text`	sst2	8	0.621±0.043	0.621±0.043	1.000	0	1	1.295±0.084	0.086±0.046

Table 2: Simulability certificate for all models. Mean and standard deviation over seeds. Acc@χ=1 is the accuracy of a product-state surrogate; $p^{\chi=1}_{\mathrm{McN}}$ is the mean exact-McNemar p-value comparing that surrogate against the full model, where a large value means the two are not significantly different and a small value flags a surrogate that is significantly worse; $d^{\chi=1}$ is the mean number of test examples on which the surrogate and the full model disagree, which is what separates a p-value of one obtained because the two models predict identically (d=0) from one obtained because their disagreements happen to be balanced; $\chi^\star$ is Eq. 1; $\bar{S}$ is the mean bipartite entanglement entropy in nats; ΔA_{ent} is the entanglement-removal gap, marked `n/a` where a separable twin is not defined. Attention and mixer models are recovered at χ=1 with no disagreements at all, and `discocat` needs χ=2.

Reproduction, honestly. Table 3 and Figure 9 place our reimplementations against the published numbers. On MC the compositional model reproduces to within 0.02. On RP, a task of only 74 training items that the original papers themselves flag as fragile, our numbers run about 0.14 to 0.18 below published, which we attribute to our own token embeddings and, for DisCoCat, the offline reader standing in for the unavailable grammar parser. That shortfall is the mean of a wide distribution rather than an unreachable target. Over 20 seeds each model's RP accuracy spans roughly 0.36 to 0.74, and the published value lies inside that spread for the compositional model and for `qsann`, and 0.014 above our best seed for `qmsan`. On a 31-example test set one example is 3.2 accuracy points, so RP cannot resolve differences of that size; we therefore report the seed mean throughout, since a best-of-seeds figure compared against a published point

Model	Data	seeds	ours (test)	published
`discocat`	mc_real	3	0.778±0.087	0.798
`discocat`	rp	20	0.565±0.098	0.723
`qmsan`	rp	20	0.579±0.063	0.756
`qsann`	rp	20	0.539±0.085	0.677
`vqc_text`	sst2	8	0.621±0.043	n/a

Table 3: Reproduction gap: our audited accuracy against the published number, on the datasets with a published value. MC_REAL reproduces to within 0.02, whereas RP runs below published, being a fragile task of only 74 training items reimplemented with our own embeddings and, for DisCoCat, an offline reader. The simulability verdict concerns our own trained circuits and is independent of this gap.

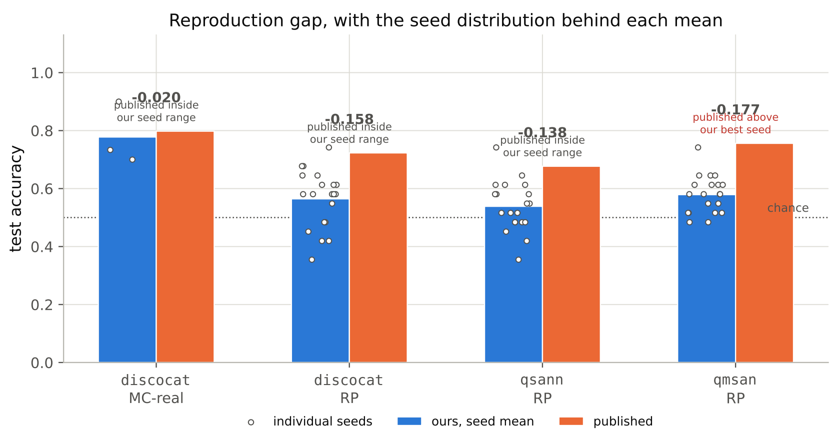


Figure 9: Reproduction gap, with the seed distribution behind each mean. Bars are our seed mean against the published number; the bold annotation is ours minus published, and each open circle is one seed. MC-REAL reproduces to within 0.02. On RP, a fragile task of only 74 training items reimplemented with our own embeddings and an offline reader, the mean runs below published, but the seed spread is wide enough that the published value falls inside our range for `discocat` and `qsann`, and 0.014 above our best seed for `qmsan`. Seed means and counts here are produced by the same selection rule as Table 3, so the figure and the table report the same quantity. The simulability verdict concerns our own trained circuits and does not depend on closing this gap.

estimate would not be a like-for-like comparison. We flag this plainly, because it matters for one thing and not another. It weakens any claim about matching RP accuracy, but the simulability verdict is about our own trained circuits and is unaffected by whether they hit the published accuracy. The certificate audits the model in hand. One might still object that better-trained models would demand more bond dimension, so that our gap, rather than any real property, drives the verdict. The evidence cuts the other way. On MC, where we reproduce the published accuracy, the attention models are already recovered at χ=1, so the χ=1 finding does not depend on the RP shortfall; and $\chi^\star$ is flat in system size on a task we do fit (Figure 5), which is the opposite of what a training-quality artefact would produce.

8 Discussion, caveats, and open questions

The scoped claim is narrow and, we think, well supported. The QNLP models we audited reproduce their behavior under a classical surrogate of small and non-growing bond dimension, and the entanglement they carry is, with one revealing exception, not load-bearing for their predictions, at the model sizes and datasets tested. That scope is a real limit rather than a hedge, since small-scale simulation cannot reach the asymptotic question, but what it does reach it answers cleanly.

Several caveats bound the reading. The audit is average-case over a test set, not a statement about every input. $\chi^\star$ is defined against a tolerance, and while we check robustness to its choice, a different tolerance shifts the borderline cases. Our simulability notion is the practical one, up to data loading, not a hardness result. The scales are small, and although the point of the scaling study is that they need not be large for the pattern to appear, a determined optimum at larger n remains out of simulation's reach. Our reimplementations also fall short of the published accuracies on RP, so for those models we audit weaker instances than the originals, though the verdict concerns the circuits we actually trained rather than the published ones. Finally, the all-quantum mixer runs at reduced scale on a CPU, where its accuracy is near chance, so we treat it as a demonstration that the audit works at eight qubits rather than as an audit of the published model. That model remains untested at its full scale, which would need a GPU to reproduce, and we say so plainly rather than presenting a reduced-scale run as a full audit.[3]

There is a constructive reading. If these models are bond-cheap at deployment scale, their quantum implementation earns its overhead only where a classical MPS cannot follow, that is where $\chi^\star$ grows with system size, and none of the audited models show it growing. The audit therefore names the target a genuine quantum ingredient must hit rather than merely cataloguing a shortfall. That reframing turns the negative headline into a design target and suggests concrete questions for QNLP. Which architectural choices, if any, make $\chi^\star$ grow with n? Do the compositional cups that make `discocat` need $\chi=2$ point toward structure worth scaling? Should QNLP benchmarks report a simulability certificate alongside accuracy, the way tabular QML increasingly does? Our position is collegial rather than combative, since the tabular and vision audits reached the same conclusion in their domains (Bowles et al., 2024; Bermejo et al., 2024), and we read our results as extending, not contradicting, theirs. *Until a QNLP model shows its bond dimension growing, the burden of proof for a quantum benefit sits with the model, not with the skeptic.*

Reproducibility statement

The harness, all model reimplementations, exact configurations, seeds, dataset indices, and the scripts that regenerate every figure and table from stored results are released at the anonymized repository above. The audit is deterministic given a trained model, and the only stochasticity is in training, which we control by seed.

References

A. Abbas, D. Sutter, C. Zoufal, A. Lucchi, A. Figalli, and S. Woerner. The power of quantum neural networks. *Nature Computational Science* 1:403-409, 2021.

E. R. Anschuetz and B. T. Kiani. Quantum variational algorithms are swamped with traps. *Nature Communications* 13:7760, 2022. arXiv:2205.05786.

M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology* 4:043001, 2019. arXiv:1906.07682.

V. Bergholm, J. Izaac, M. Schuld, C. Gogolin, and N. Killoran. PennyLane: automatic differentiation of hybrid quantum-classical computations. *arXiv:1811.04968*, 2018.

P. Bermejo et al. Quantum convolutional neural networks are (effectively) classically simulable. *arXiv:2408.12739*, 2024.

J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd. Quantum machine learning. *Nature* 549:195-202, 2017. arXiv:1611.09347.

J. Bowles, S. Ahmed, and M. Schuld. Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv:2403.07059*, 2024.

M. C. Caro, H.-Y. Huang, M. Cerezo, K. Sharma, A. Sornborger, L. Cincio, and P. J. Coles. Generalization in quantum machine learning from few training data. *Nature Communications* 13:4919, 2022. arXiv:2111.05292.

[3]This is the only place additional compute would change what we can measure; the rest of the audit is CPU-bound by design.

M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. Variational quantum algorithms. *Nature Reviews Physics* 3:625-644, 2021.

M. Cerezo et al. Does provable absence of barren plateaus imply classical simulability? *Nature Communications*, 2025. arXiv:2312.09121.

F. Chen et al. Quantum mixed-state self-attention network. *arXiv:2403.02871*, 2024.

J. Chen et al. CLAQS: compact learnable all-quantum token mixer for text classification. *arXiv:2510.06532*, 2025.

B. Coecke, M. Sadrzadeh, and S. Clark. Mathematical foundations for a compositional distributional model of meaning. *Linguistic Analysis* 36:345-384, 2010. arXiv:1003.4394.

I. Cong, S. Choi, and M. D. Lukin. Quantum convolutional neural networks. *Nature Physics* 15:1273-1278, 2019. arXiv:1810.03787.

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171-4186, 2019.

T. G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation* 10(7):1895-1923, 1998.

T. Duneau, S. Bruhn, G. Matos, T. Laakkonen, K. Saiti, A. Pearson, K. Meichanetzidis, and B. Coecke. Scalable and interpretable quantum natural language processing: an implementation on trapped ions. *arXiv:2409.08777*, 2024.

B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics* 7(1):1-26, 1979.

V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature* 567:209-212, 2019.

H.-Y. Huang, R. Kueng, and J. Preskill. Predicting many properties of a quantum system from very few measurements. *Nature Physics* 16:1050-1057, 2020.

H.-Y. Huang et al. Power of data in quantum machine learning. *Nature Communications* 12:2631, 2021. arXiv:2011.01938.

D. Kartsaklis, I. Fan, R. Yeung, A. Pearson, R. Lorenz, A. Toumi, G. de Felice, K. Meichanetzidis, S. Clark, and B. Coecke. lambeq: an efficient high-level Python library for quantum NLP. *arXiv:2110.04236*, 2021.

M. Larocca et al. Barren plateaus in variational quantum computing. *Nature Reviews Physics*, 2025. doi:10.1038/s42254-025-00813-9.

G. Li, X. Zhao, and X. Wang. Quantum self-attention neural networks for text classification. *arXiv:2205.05625*, 2022.

Y. Liu, S. Arunachalam, and K. Temme. A rigorous and robust quantum speed-up in supervised machine learning. *Nature Physics* 17:1013-1017, 2021.

R. Lorenz et al. QNLP in practice: running compositional models of meaning on a quantum computer. *arXiv:2102.12846*, 2021.

J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications* 9:4812, 2018.

Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12:153-157, 1947.

K. Meichanetzidis, S. Gogioso, G. de Felice, N. Chiappori, A. Toumi, and B. Coecke. Quantum natural language processing on near-term quantum computers. *Electronic Proceedings in Theoretical Computer Science* 340:213-229, 2021. arXiv:2005.04147.

K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii. Quantum circuit learning. *Physical Review A* 98:032309, 2018.

F. Nausheen, K. Ahmed, M. I. Khan, and F. Riaz. Quantum natural language processing: a comprehensive review of models, methods, and applications. *arXiv:2504.09909*, 2025.

R. Orús. A practical introduction to tensor networks: matrix product states and projected entangled pair states. *Annals of Physics* 349:117-158, 2014. arXiv:1306.2164.

A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O'Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications* 5:4213, 2014.

A. Pesah, M. Cerezo, S. Wang, T. Volkoff, A. T. Sornborger, and P. J. Coles. Absence of barren plateaus in quantum convolutional neural networks. *Physical Review X* 11:041011, 2021.

J. Preskill. Quantum computing in the NISQ era and beyond. *Quantum* 2:79, 2018.

U. Schollwöck. The density-matrix renormalization group in the age of matrix product states. *Annals of Physics* 326(1):96-192, 2011. arXiv:1008.3477.

F. J. Schreiber, J. Eisert, and J. J. Meyer. Classical surrogates for quantum learning models. *Physical Review Letters* 131:100803, 2023. arXiv:2206.11740.

M. Schuld and N. Killoran. Quantum machine learning in feature Hilbert spaces. *Physical Review Letters* 122:040504, 2019. arXiv:1803.07128.

M. Schuld, R. Sweke, and J. J. Meyer. Effect of data encoding on the expressive power of variational quantum machine learning models. *Physical Review A* 103:032430, 2021.

M. Schuld. Supervised quantum machine learning models are kernel methods. *arXiv:2101.11020*, 2021.

S. Shin, Y. S. Teo, and H. Jeong. Dequantizing quantum machine learning models using tensor networks. *Physical Review Research* 6:023218, 2024.

S. Sim, P. D. Johnson, and A. Aspuru-Guzik. Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms. *Advanced Quantum Technologies* 2:1900070, 2019. arXiv:1905.10876.

R. Socher, A. Perelygin, J. Wu, J. Chuang, C. D. Manning, A. Ng, and C. Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of EMNLP*, pages 1631-1642, 2013.

E. M. Stoudenmire and D. J. Schwab. Supervised learning with tensor networks. In *Advances in Neural Information Processing Systems 29*, 2016. arXiv:1605.05775.

E. Tang. A quantum-inspired classical algorithm for recommendation systems. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, pages 217-228, 2019. arXiv:1807.04271.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30*, pages 5998-6008, 2017.

F. Verstraete and J. I. Cirac. Matrix product states represent ground states faithfully. *Physical Review B* 73:094423, 2006. arXiv:cond-mat/0505140.

G. Vidal. Efficient classical simulation of slightly entangled quantum computations. *Physical Review Letters* 91:147902, 2003. arXiv:quant-ph/0301063.

S. R. White. Density matrix formulation for quantum renormalization groups. *Physical Review Letters* 69(19):2863-2866, 1992.

W. Zeng and B. Coecke. Quantum algorithms for compositional natural language processing. *Electronic Proceedings in Theoretical Computer Science* 221:67-75, 2016. arXiv:1608.01406.

A Literature-review methodology

The motivating claim in Section 1, that QNLP text-classification papers argue a quantum benefit from accuracy but rarely test whether the *trained circuit* is classically simulable, rests on a focused review rather than an exhaustive systematic survey, and we scope it honestly here.

Search and inclusion. We searched arXiv (`quant-ph`, `cs.CL`) and Semantic Scholar for the conjunctions of {"quantum"} with {"natural language processing", "text classification", "self-attention", "sentence classification"} over 2019-2025, and followed citations from the models we reimplement. We include a paper if it (i) proposes a quantum or hybrid model for a text-classification task and (ii) reports an accuracy comparison. We record, for each, whether it reports a *classical surrogate of the trained circuit*, meaning a tensor-network contraction, a Pauli-propagation estimate, or an explicit entanglement-removal ablation, as opposed to merely a separately-trained classical network of comparable size.

What we found. Table 4 lists the core set. All report a classical *network* baseline; none report a classical surrogate of their own trained circuit, and none report the bond dimension, entanglement, or a simulability check of the trained model. This is the gap SimCert fills. We stress the limits: this is a focused review of the papers central to this study plus those surfaced by the searches above, not a saturated systematic survey, and we did not attempt inter-rater agreement. We therefore phrase the claim in the main text as scoped to "the papers we examined," and treat a full systematic survey (with pre-registered search strings and dual coding, in the style of the tabular-QML audit of Bowles et al., 2024) as future work.

Paper	classical-network baseline	trained-circuit surrogate	reports χ / entanglement
DisCoCat / `lambeq` (Lorenz et al., 2021)	- [a]	no	no
`QSANN` (Li et al., 2022)	yes	no	no
`QMSAN` (Chen et al., 2024)	yes	no	no
`CLAQS` (Chen et al., 2025)	yes	no	no

Table 4: Core reviewed set. [a]DisCoCat compares to a bag-of-words/word-sequence baseline rather than a matched network. None reports a classical surrogate of the trained circuit or a simulability diagnostic.

B Model circuits and glossary

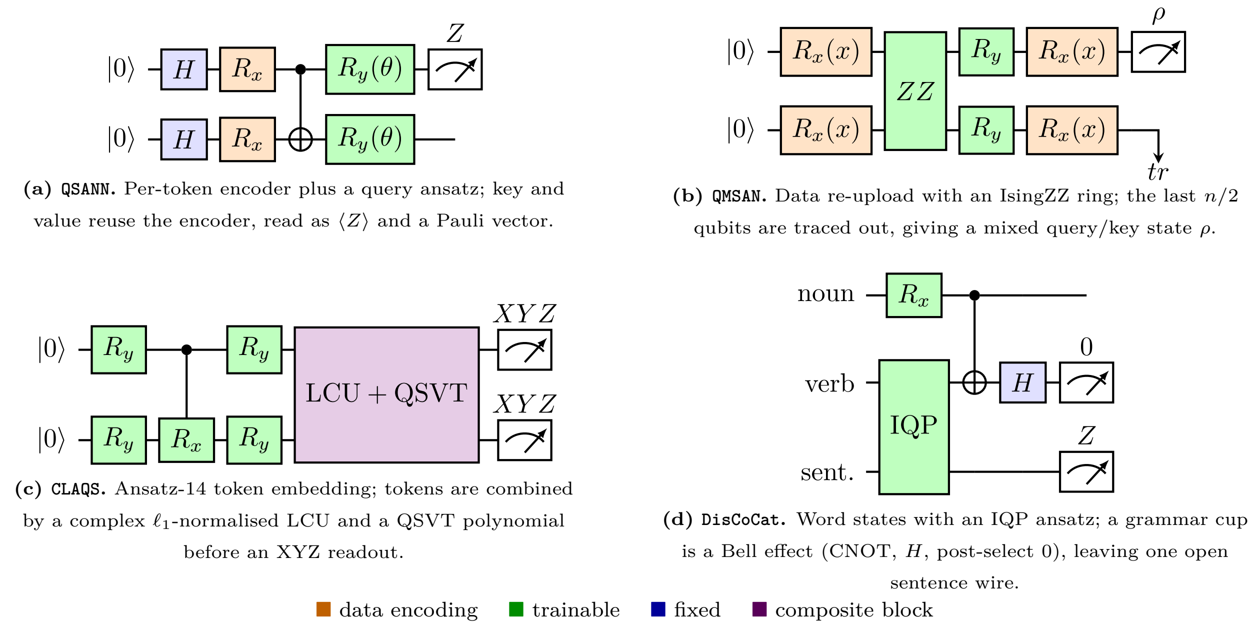
All models are reimplemented in PennyLane; `lambeq` produces the DisCoCat circuits. We give each model's trained state below, and Figure 10 shows a representative circuit for each; the audit truncates that state (Algorithm 1).

`vqc_text` (reference). Bag-of-embeddings feature $x \in \mathbb{R}^n$ angle-encoded on n qubits, L layers of $R_y(x_i)$ (re-uploaded), trainable R_y, R_z, and a CNOT ring; readout $\langle Z_0 \rangle$ (Fig. 3).

`QSANN` (Gaussian-projected quantum self-attention). Per token, the same strongly-entangling template ($H^{\otimes n}$, then R_x/R_y layers with CNOT chains) prepares $|\psi_s\rangle = U_{enc}(x_s)H^{\otimes n}|0\rangle$; three trainable ansätze U_q, U_k, U_v give a query scalar $\langle Z_0 \rangle$, a key scalar, and a d-dimensional value vector of single-qubit Pauli expectations. Attention $\alpha_{s,j} = \exp(-(\langle Z_q \rangle_s - \langle Z_k \rangle_j)^2)$, the residual sum, mean-pool, and sigmoid are all classical. The audit therefore acts *per token* on the pure states $U_{q,k,v}|\psi_s\rangle$, and the classical head is the composition function.

`QMSAN` (mixed-state self-attention). Per token, a data-reuploading circuit ($R_x(x_i)$; $L \times$ [IsingZZ ring, R_y layer]; re-upload) prepares a pure state whose reduced density matrix on the first $n/2$ qubits is the query/key object; attention is the Hilbert-Schmidt overlap $\alpha_{s,j} = \mathrm{tr}(\rho_{q,s}\sigma_{k,j})$. We audit this on the *purification*: we truncate the pure n-qubit state to bond dimension χ, then take its partial trace to form ρ, σ. Values are per-qubit $\langle Z_i \rangle$.

`CLAQS` (all-quantum token mixer). On q=8 data qubits, each token's ansatz-14 embedding unitary U_j is combined by a learnable, ℓ_1-normalised complex linear-combination-of-unitaries $M = \sum_j \bar{b}_j U_j$; a learnable QSVT polynomial $P_c(M) = \sum_{k=0}^{5} c_k M^k$ and a window feed-forward unitary U_{FF} produce $|\psi\rangle =$ normalise($U_{\mathrm{FF}}\, P_c(M)\, |0^8\rangle$); readout is 24 XYZ Pauli expectations into a two-layer MLP. Because $|\psi\rangle$ is a matrix polynomial rather than a gate sequence, the audit consumes the trained statevector directly. At q=8 the exact MPS bond dimension is 16, which is why this is the one model where the χ-axis has room to resolve structure.



(a) **`QSANN`.** Per-token encoder plus a query ansatz; key and value reuse the encoder, read as $\langle Z \rangle$ and a Pauli vector.

(b) **`QMSAN`.** Data re-upload with an IsingZZ ring; the last $n/2$ qubits are traced out, giving a mixed query/key state ρ.

(c) **`CLAQS`.** Ansatz-14 token embedding; tokens are combined by a complex ℓ_1-normalised LCU and a QSVT polynomial before an XYZ readout.

(d) **`DisCoCat`.** Word states with an IQP ansatz; a grammar cup is a Bell effect (CNOT, H, post-select 0), leaving one open sentence wire.

Figure 10: Representative circuits for the four audited models (simplified; exact per-model circuits are released with the harness). Gates are colored by role. `QMSAN` is read out as a reduced density matrix, hence the mixed-state audit path; `DisCoCat` post-selects the cup qubits and reads the single open wire by the Born rule.

`discocat`. A sentence's pregroup structure becomes one circuit: single-qubit word states (R_x / Euler $R_xR_zR_x$), IQP ansätze for multi-qubit words (per layer: H on all wires, then a chain of adjacent CR_z), a parameter-free GHZ for the relative pronoun, and grammar cups realised as Bell effects (post-select the paired qubits on $|0\rangle$). We audit the whole-sentence pure state before post-selection, then post-select and read the open wire's Born rule.

Glossary (for NLP readers). **NISQ**: noisy intermediate-scale quantum hardware. **PQC**: parameterised quantum circuit. **Ansatz**: a fixed circuit template whose rotation angles are trainable. R_x, R_y, R_z: single-qubit rotations; **CNOT**/CR_z: two-qubit entangling gates. **Statevector**: the 2^n complex amplitudes of an n-qubit pure state. **MPS** (matrix product state): a tensor-network factorisation whose **bond dimension** χ caps the entanglement it can represent; χ=1 is a product (unentangled) state. **Entanglement entropy** $\bar{S}$: bipartite von Neumann entropy of the state (nats). **IQP / LCU / QSVT**: instantaneous-quantum-polynomial ansatz / linear combination of unitaries / quantum singular-value transformation. **CSIM/QESIM**: classically simulable / classically simulable up to data loading.

C Datasets and preprocessing

MC (meaning classification). We use two variants. MC-REAL is the Lorenz et al. set (70 train / 30 dev / 30 test; 17-word vocabulary; 3-4 word cooking-vs-computing sentences), obtained from the authors' public resources repository; POS tags are stripped (`word_N`→`word`) since our bag-of-token models do not use them and the DisCoCat reader re-parses. MC-SYNTH is a self-contained stand-in generated from disjoint food/IT subject-verb-object templates (65 per class), used only for the controlled scaling study; being disjoint-vocabulary it is near-linearly-separable, which is why every model saturates on it.

RP (relative pronoun). The Lorenz et al. RELPRON-derived set (74 train / 31 test; 115-word vocabulary; four-token noun phrases), distinguishing subject- from object-relative clauses. We carve a 20% validation split from train. RP is the discriminating benchmark and, with only 74 training items, the fragile one, and the original papers note near-degenerate behaviour here, which we reproduce.

SST-2. A balanced subset of the Stanford Sentiment Treebank binary task (600 train / 200 test, drawn from the train and labelled-validation splits respectively via the HuggingFace datasets server), sequence length capped at 16 tokens. A genuinely non-separable, real-text task; accuracies here are in the 0.5-0.65 range, not saturated.

Embeddings and tokenisation. For `qsann`, `qmsan`, `claqs`, and `vqc_text` we use a trainable per-token embedding (dimension equal to the model's feature width); `vqc_text` averages token embeddings (bag-of-embeddings), the attention models keep per-token vectors, and `claqs` maps each token embedding through a trainable linear layer to ansatz-14 angles. DisCoCat does not use embeddings: each word's parameters *are* its gate angles, trained via the `lambeq` pipeline with the offline `cups_reader` (the trained Bobcat grammar parser's model host is no longer reachable, an acknowledged fidelity cost on RP). Splits are seed-stable; the exact indices and the synthetic generator are released with the harness so every split is reproducible.

D Configurations, compute, and robustness

Hyperparameters. Table 5 lists the per-model settings. Quantum models train with Adam (AdamW for `claqs`); `discocat` trains with SPSA (a=0.05, c=0.06) as in the original pipeline. Quantum-model weights are initialised $\mathcal{N}(0, 0.01)$; all runs use seeds $\{1, 2, 3\}$. The matched classical baseline is logistic regression on bag-of-words counts. Quantum-model feature widths follow the source papers (n=2 qubits on MC, n=4 on RP; q=8 for `claqs`).

Model	qubits	depth/layers	optimiser	epochs
`vqc_text`	4-10	2	Adam (0.05)	30-40
`qsann`	2 / 4	D_{enc}=1, D_{qkv}=1	Adam (0.05)	40
`qmsan`	2 / 4	L=1 (IsingZZ ring)	Adam (0.05)	40
`claqs`	8	L=1, QSVT degree 5	AdamW (0.05)	10-12
`discocat`	3-7	IQP L=1	SPSA	120

Table 5: Per-model training configuration. Full configs and the exact commands are released with the harness.

Compute. All experiments run on a single Apple M1 CPU (8 GB RAM, no GPU). The audit is deterministic given a trained model, since expectation values are computed exactly rather than sampled, so the only stochasticity is training. Statevector simulation is memory-light at these sizes (a 16-qubit state is ≈ 1 MB); the binding constraint is training the attention models, not the audit. `claqs` is the exception: a faithful full-scale reproduction of its published 91.6% on SST-2 needs a GPU, so we run it at reduced scale (100 training examples, $\leq$ 6-token windows, 10 epochs); its SST-2 accuracy is consequently near chance and we report the audit, not an accuracy reproduction.

Robustness of $\chi^\star$ to the tolerance. Equation 1 defines $\chi^\star$ against a tolerance τ. We compute it under three definitions: τ_{gen} (the model's train-test generalization gap, our default), τ_{ci} (the half-width of a 95% bootstrap CI on full-χ accuracy), and τ_{agree} (smallest χ with prediction agreement $\pi(\chi) \geq 0.95$). Across all runs the three agree on $\chi^\star$ in the large majority of cases, namely χ=1 for the attention and mixer models and χ=2 for `discocat`, and where they differ they do so by a single bond-dimension step on borderline, low-accuracy RP runs. The headline conclusion is therefore not an artefact of the tolerance choice.